Experiment 2

Title: Apply Various Text Preprocessing Techniques for a Given Text (Tokenization & Filtration, Script Validation)

Aim

To apply different **text preprocessing techniques** such as tokenization, filtration, and script validation on a given text dataset.

Objectives

1. To understand the concept of text preprocessing in data analysis.
2. To apply tokenization and filtration techniques on text data.
3. To study script validation methods used in text processing.

Theory

1. Introduction to Text Preprocessing
Text preprocessing is an important step in **Natural Language Processing (NLP)** and text analysis. Raw text data often contains noise such as punctuation, extra spaces, and irrelevant words. Preprocessing helps in cleaning and preparing the text so that it can be analyzed effectively.

2. Tokenization
Tokenization is the process of **breaking a large piece of text into smaller units called tokens**. These tokens can be words, sentences, or characters depending on the requirement. Tokenization simplifies the text and makes it easier for algorithms to process and analyze the data.

3. Filtration in Text Processing
Filtration is used to remove **unnecessary or irrelevant elements** from the text. This may include removing punctuation marks, special characters, numbers, and common stop words such as “the”, “is”, and “and”. Filtration improves the quality of the dataset and reduces noise during analysis.

4. Script Validation
Script validation is the process of checking whether the text belongs to a **specific writing script or language**. It ensures that the text data follows the required format and language before further processing. This step helps maintain consistency and accuracy in text analysis tasks.

Conclusion

In this experiment, different text preprocessing techniques such as tokenization, filtration, and script validation were studied and applied. These techniques help in cleaning and structuring raw text data before analysis. Proper preprocessing improves the quality and reliability of the dataset. It also helps machine learning models understand the text more effectively. Therefore, text preprocessing is a crucial step in natural language processing tasks.